INCH POUND

MIL-C-3098/100B
5 May 1993
SUPERSEDING
MIL-C-3098/100A
2 November 1983

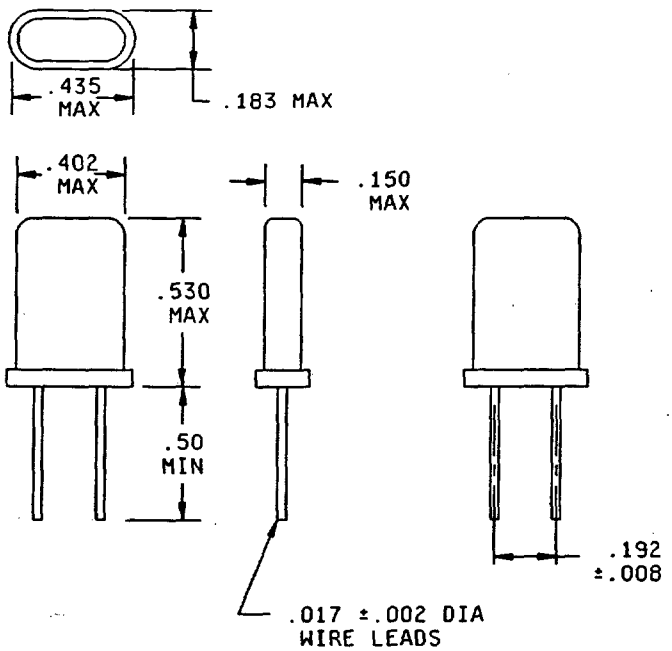
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR123/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth mechanical overtone; controlled; series-resonance.




| Inches | mm |
|---|---|
| .002 | .05 |
| .008 | .20 |
| .017 | .43 |
| .150 | 3.81 |
| .183 | 4.65 |
| .192 | 4.88 |
| .402 | 10.21 |
| .435 | 11.05 |
| .50 | 12.7 |
| .530 | 13.46 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR123/U.


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REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/9) or HC-43/U.

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50 to 125 MHz.

Capacitance, shunt: 4.5 pF, maximum.

Frequency tolerances:

Operating temperature range: ±10 ppm.

Reference temperature: ±10 ppm.

Room temperature: ±80 ppm.

Frequency stability: ±2.5 ppm.

Mode of oscillation: Fifth mechanical overtone.

Rated drive level: 2.0 ±0.4 mW.

Resonance: Series.

Reference temperature: +85°C ±1°C.

Temperature ranges:

Operating (controlled): +80°C to +90°C, inclusive.

Operable: -55°C to +80°C and +90°C to +100°C.

Equivalent resistance: 60 ohms, maximum.

Unwanted modes: Minimum unwanted mode effective resistance of 120 ohms or an unwanted to main mode resistance ratio of 3:1, whichever is greater.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A, 0.06 inch double amplitude or 5 gravity units, whichever is less.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock: ±10 ppm.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.



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CONCLUDING MATERIAL

Custodians:

Army - ER
Navy - EC
Air Force - 85

User activities:

Air Force - 17, 19

Preparing activity:

Army - ER

Agent:

DLA - ES

(Project 5955-0678-15)



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